

Source Code:

// generation & propagation

```
module gp (in_a, in_b, out_g, out_p);
    input in_a, in_b;
    output out_g, out_p;

    assign out_g = in_a & in_b;
    assign out_p = in_a ^ in_b;
endmodule
```

// get g(i:k) & p(i:k)

```
module merge (in_ga, in_pa, in_gb, in_pb, out_gnew, out_pnew);
    input in_ga, in_pa, in_gb, in_pb;
    output out_gnew, out_pnew;

    assign out_gnew = in_gb | (in_pb & in_ga);
    assign out_pnew = in_pb & in_pa;
endmodule
```

endmodule

// summation module

```
module sum (in_p, in_cin, out_s);
    input in_p, in_cin;
    output out_s;

    assign out_s = in_p ^ in_cin;
    //assign out_cout = in_g | (in_p & in_cin);
endmodule
```

// carry out module

```
module carry (in_g, in_p, in_cin, out_cout);
    input in_g, in_p, in_cin;
    output out_cout;

    //assign out_s = in_p ^ in_cin;
    assign out_cout = in_g | (in_p & in_cin);
endmodule
```

//prefix adder KSA

```
module ksa10 (i_a, i_b, i_cin, o_s, o_cout);
    input [9:0] i_a, i_b;
    input i_cin;
    output [9:0] o_s;
    output o_cout;
```

// generate g & p

```
wire w_g0, w_g1, w_g2, w_g3, w_g4, w_g5, w_g6, w_g7, w_g8, w_g9;
wire w_p0, w_p1, w_p2, w_p3, w_p4, w_p5, w_p6, w_p7, w_p8, w_p9;
```

```
gp A0 (.in_a(i_a[0]), .in_b(i_b[0]), .out_g(w_g0), .out_p(w_p0));
gp A1 (.in_a(i_a[1]), .in_b(i_b[1]), .out_g(w_g1), .out_p(w_p1));
gp A2 (.in_a(i_a[2]), .in_b(i_b[2]), .out_g(w_g2), .out_p(w_p2));
gp A3 (.in_a(i_a[3]), .in_b(i_b[3]), .out_g(w_g3), .out_p(w_p3));
gp A4 (.in_a(i_a[4]), .in_b(i_b[4]), .out_g(w_g4), .out_p(w_p4));
gp A5 (.in_a(i_a[5]), .in_b(i_b[5]), .out_g(w_g5), .out_p(w_p5));
gp A6 (.in_a(i_a[6]), .in_b(i_b[6]), .out_g(w_g6), .out_p(w_p6));
gp A7 (.in_a(i_a[7]), .in_b(i_b[7]), .out_g(w_g7), .out_p(w_p7));
gp A8 (.in_a(i_a[8]), .in_b(i_b[8]), .out_g(w_g8), .out_p(w_p8));
gp A9 (.in_a(i_a[9]), .in_b(i_b[9]), .out_g(w_g9), .out_p(w_p9));
```

// generate gnew

```
wire w_gn0, w_gn1, w_gn2, w_gn3, w_gn4, w_gn5, w_gn6, w_gn7, w_gn8, w_gn9, w_gn10, w_gn11, w_gn12, w_gn13, w_gn14,
w_gn15, w_gn16, w_gn17, w_gn18, w_gn19, w_gn20, w_gn21, w_gn22, w_gn23, w_gn24;
wire w_pn0, w_pn1, w_pn2, w_pn3, w_pn4, w_pn5, w_pn6, w_pn7, w_pn8, w_pn9, w_pn10, w_pn11, w_pn12, w_pn13, w_pn14,
w_pn15, w_pn16, w_pn17, w_pn18, w_pn19, w_pn20, w_pn21, w_pn22, w_pn23, w_pn24;
```

//first layer

```
merge U0 (.in_ga(w_g0), .in_pa(w_p0), .in_gb(w_g1), .in_pb(w_p1), .out_gnew(w_gn0), .out_pnew(w_pn0)); //g,p(1:0)
merge U1 (.in_ga(w_g1), .in_pa(w_p1), .in_gb(w_g2), .in_pb(w_p2), .out_gnew(w_gn1), .out_pnew(w_pn1)); //g,p(2:1)
merge U2 (.in_ga(w_g2), .in_pa(w_p2), .in_gb(w_g3), .in_pb(w_p3), .out_gnew(w_gn2), .out_pnew(w_pn2)); //g,p(3:2)
merge U3 (.in_ga(w_g3), .in_pa(w_p3), .in_gb(w_g4), .in_pb(w_p4), .out_gnew(w_gn3), .out_pnew(w_pn3)); //g,p(4:3)
merge U4 (.in_ga(w_g4), .in_pa(w_p4), .in_gb(w_g5), .in_pb(w_p5), .out_gnew(w_gn4), .out_pnew(w_pn4)); //g,p(5:4)
```

```

merge U5 (.in_ga(w_g5), .in_pa(w_p5), .in_gb(w_g6), .in_pb(w_p6), .out_gnew(w_gn5), .out_pnew(w_pn5)); //g,p(6:5)
merge U6 (.in_ga(w_g6), .in_pa(w_p6), .in_gb(w_g7), .in_pb(w_p7), .out_gnew(w_gn6), .out_pnew(w_pn6)); //g,p(7:6)
merge U7 (.in_ga(w_g7), .in_pa(w_p7), .in_gb(w_g8), .in_pb(w_p8), .out_gnew(w_gn7), .out_pnew(w_pn7)); //g,p(8:7)
merge U8 (.in_ga(w_g8), .in_pa(w_p8), .in_gb(w_g9), .in_pb(w_p9), .out_gnew(w_gn8), .out_pnew(w_pn8)); //g,p(9:8)
//second layer
merge U9 (.in_ga(w_g0), .in_pa(w_p0), .in_gb(w_gn1), .in_pb(w_pn1), .out_gnew(w_gn9), .out_pnew(w_pn9)); //g,p(2:0)
merge U10 (.in_ga(w_gn0), .in_pa(w_pn0), .in_gb(w_gn2), .in_pb(w_pn2), .out_gnew(w_gn10), .out_pnew(w_pn10)); //g,p(3:0)
merge U11 (.in_ga(w_gn1), .in_pa(w_pn1), .in_gb(w_gn3), .in_pb(w_pn3), .out_gnew(w_gn11), .out_pnew(w_pn11)); //g,p(4:1)
merge U12 (.in_ga(w_gn2), .in_pa(w_pn2), .in_gb(w_gn4), .in_pb(w_pn4), .out_gnew(w_gn12), .out_pnew(w_pn12)); //g,p(5:2)
merge U13 (.in_ga(w_gn3), .in_pa(w_pn3), .in_gb(w_gn5), .in_pb(w_pn5), .out_gnew(w_gn13), .out_pnew(w_pn13)); //g,p(6:3)
merge U14 (.in_ga(w_gn4), .in_pa(w_pn4), .in_gb(w_gn6), .in_pb(w_pn6), .out_gnew(w_gn14), .out_pnew(w_pn14)); //g,p(7:4)
merge U15 (.in_ga(w_gn5), .in_pa(w_pn5), .in_gb(w_gn7), .in_pb(w_pn7), .out_gnew(w_gn15), .out_pnew(w_pn15)); //g,p(8:5)
merge U16 (.in_ga(w_gn6), .in_pa(w_pn6), .in_gb(w_gn8), .in_pb(w_pn8), .out_gnew(w_gn16), .out_pnew(w_pn16)); //g,p(9:6)
//third layer
merge U17 (.in_ga(w_g0), .in_pa(w_p0), .in_gb(w_gn11), .in_pb(w_pn11), .out_gnew(w_gn17), .out_pnew(w_pn17)); //g,p(4:0)
merge U18 (.in_ga(w_gn0), .in_pa(w_pn0), .in_gb(w_gn12), .in_pb(w_pn12), .out_gnew(w_gn18), .out_pnew(w_pn18)); //g,p(5:0)
merge U19 (.in_ga(w_gn9), .in_pa(w_pn9), .in_gb(w_gn13), .in_pb(w_pn13), .out_gnew(w_gn19), .out_pnew(w_pn19)); //g,p(6:0)
merge U20 (.in_ga(w_gn10), .in_pa(w_pn10), .in_gb(w_gn14), .in_pb(w_pn14), .out_gnew(w_gn20), .out_pnew(w_pn20)); //g,p(7:0)
merge U21 (.in_ga(w_gn11), .in_pa(w_pn11), .in_gb(w_gn15), .in_pb(w_pn15), .out_gnew(w_gn21), .out_pnew(w_pn21)); //g,p(8:1)
merge U22 (.in_ga(w_gn12), .in_pa(w_pn12), .in_gb(w_gn16), .in_pb(w_pn16), .out_gnew(w_gn22), .out_pnew(w_pn22)); //g,p(9:2)
//fourth layer
merge U23 (.in_ga(w_g0), .in_pa(w_p0), .in_gb(w_gn21), .in_pb(w_pn21), .out_gnew(w_gn23), .out_pnew(w_pn23)); //g,p(8:0)
merge U24 (.in_ga(w_gn0), .in_pa(w_pn0), .in_gb(w_gn22), .in_pb(w_pn22), .out_gnew(w_gn24), .out_pnew(w_pn24)); //g,p(9:0)

// generate C
wire w_c1, w_c2, w_c3, w_c4, w_c5, w_c6, w_c7, w_c8, w_c9;
carry C0 (.in_g(w_g0), .in_p(w_p0), .in_cin(i_cin), .out_cout(w_c1));
carry C1 (.in_g(w_gn0), .in_p(w_pn0), .in_cin(i_cin), .out_cout(w_c2));
carry C2 (.in_g(w_gn9), .in_p(w_pn9), .in_cin(i_cin), .out_cout(w_c3));
carry C3 (.in_g(w_gn10), .in_p(w_pn10), .in_cin(i_cin), .out_cout(w_c4));
carry C4 (.in_g(w_gn17), .in_p(w_pn17), .in_cin(i_cin), .out_cout(w_c5));
carry C5 (.in_g(w_gn18), .in_p(w_pn18), .in_cin(i_cin), .out_cout(w_c6));
carry C6 (.in_g(w_gn19), .in_p(w_pn19), .in_cin(i_cin), .out_cout(w_c7));
carry C7 (.in_g(w_gn20), .in_p(w_pn20), .in_cin(i_cin), .out_cout(w_c8));
carry C8 (.in_g(w_gn23), .in_p(w_pn23), .in_cin(i_cin), .out_cout(w_c9));
carry C9 (.in_g(w_gn24), .in_p(w_pn24), .in_cin(i_cin), .out_cout(o_cout));

// generate S
//wire w_c1, w_c2, w_c3, w_c4, w_c5, w_c6, w_c7, w_c8, w_c9;
sum S0 (.in_p(w_p0), .in_cin(i_cin), .out_s(o_s[0]));
sum S1 (.in_p(w_p1), .in_cin(w_c1), .out_s(o_s[1]));
sum S2 (.in_p(w_p2), .in_cin(w_c2), .out_s(o_s[2]));
sum S3 (.in_p(w_p3), .in_cin(w_c3), .out_s(o_s[3]));
sum S4 (.in_p(w_p4), .in_cin(w_c4), .out_s(o_s[4]));
sum S5 (.in_p(w_p5), .in_cin(w_c5), .out_s(o_s[5]));
sum S6 (.in_p(w_p6), .in_cin(w_c6), .out_s(o_s[6]));
sum S7 (.in_p(w_p7), .in_cin(w_c7), .out_s(o_s[7]));
sum S8 (.in_p(w_p8), .in_cin(w_c8), .out_s(o_s[8]));
sum S9 (.in_p(w_p9), .in_cin(w_c9), .out_s(o_s[9]));

endmodule

`timescale 1ns/1ns
module test_top();
    reg [9:0] i_a, i_b;
    reg i_cin;

    wire [9:0] o_s;
    wire o_cout;

    // netlist
    ksa10 U0 (.i_a(i_a), .i_b(i_b), .i_cin(i_cin), .o_s(o_s), .o_cout(o_cout));

    // simulation
    initial begin
        i_a <= 10'b0000000000;
        i_b <= 10'b0000000000;
        i_cin <= 1'b0;
        //o_c:0
        //o_s:0000000000 = 000

        #10
        i_a <= 10'b0001110011;
        i_b <= 10'b1010011011;
        i_cin <= 1'b1;
    end

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```

//0
//1100001111 = 30F

#10
i_a <= 10'b1100011111;
i_b <= 10'b00111100000;
i_cin <= 1'b0;
//0
//1111111111 = 3FF

#10
i_a <= 10'b1010101010;
i_b <= 10'b0101010101;
i_cin <= 1'b1;
//1
//0000000000 = 000

#10
i_a <= 10'b1010101010;
i_b <= 10'b0101010101;
i_cin <= 1'b0;
//0
//1111111111 = 3FF

#10
i_a <= 10'b0000011111;
i_b <= 10'b0011111111;
i_cin <= 1'b1;
//0
//0100011111 = 11F

#10
i_a <= 10'b0110001101;
i_b <= 10'b00011111001;
i_cin <= 1'b0;
//0
//1000000110 = 206

#10
i_a <= 10'b1110001001;
i_b <= 10'b0011100000;
i_cin <= 1'b1;
//1
//0001101010 = 06A

#10
i_a <= 10'b1111111111;
i_b <= 10'b1111111111;
i_cin <= 1'b0;
//1
//1111111110 = 3FE

#10
i_a <= 10'b1111111111;
i_b <= 10'b1111111111;
i_cin <= 1'b1;
//1
//1111111111 = 3FF

```

end

endmodule

Simulation Result

